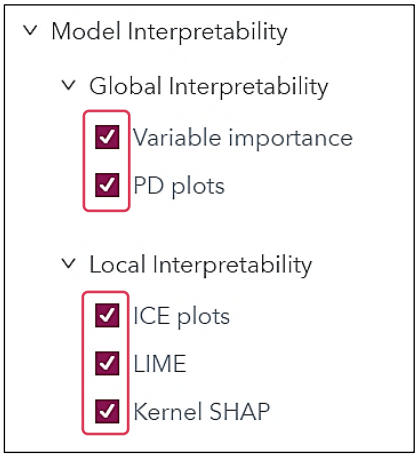
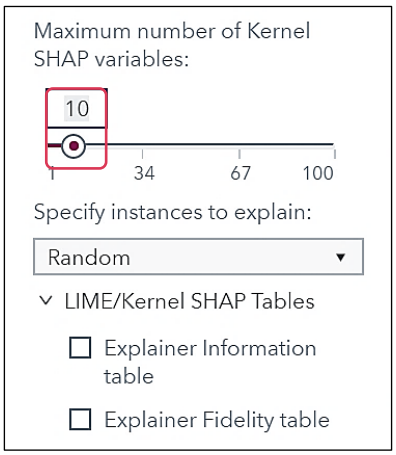
Adding Model Interpretability

In this demonstration, you use the Model Interpretability capability to add some explanation to the support vector machine model.

1. Open the **Week 7 practical lesson pipeline.** Select the SVM node, and then refer to its property panel.
2. Under **Post-training Properties**, expand **Model Interpretability**. Expand **Global Interpretability** and select the check boxes for **Variable importance** and **PD plots**. Expand **Local Interpretability** and select the check boxes for **ICE plots**, **LIME**, and **Kernel SHAP**.

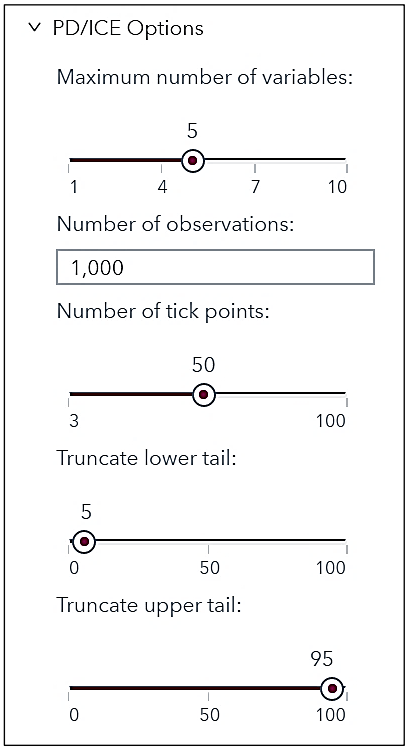


3. Scroll down in the Post-training Properties and change the maximum number of Kernel SHAP variables from 20 to **10**.



This means that 10 inputs are displayed in the chart, ordered by importance according to the absolute Kernel SHAP values. Notice that the **Random** observations option provides explanations for five randomly selected observations. However, you can specify up to five individual observations to explain for LIME, Kernel SHAP, and ICE.

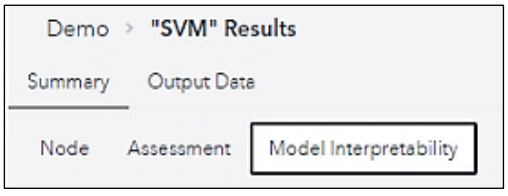
4. Scroll to the bottom of the Post-training Properties and expand **PD/ICE Options**. Explore these options. However, no change is required.



Notice that, by default, a maximum of five input variables are used to generate the PD and ICE plots. The input variables are selected based on their relative importance. The maximum number of observations to sample for the PD and ICE plots is 1000, by default.

When the input variable is interval, its extreme values are eliminated by truncating the lower and upper tails of its distribution. The amount of truncation can be controlled by the properties under **PD/ICE Options**.

5. Run the **Support Vector Machine** node.
6. Open the results for the node.
7. Now you see an additional tab along with the Node and Assessment tabs. Click the **Model Interpretability** tab.



8. Expand the **Surrogate Model Variable Importance** table.

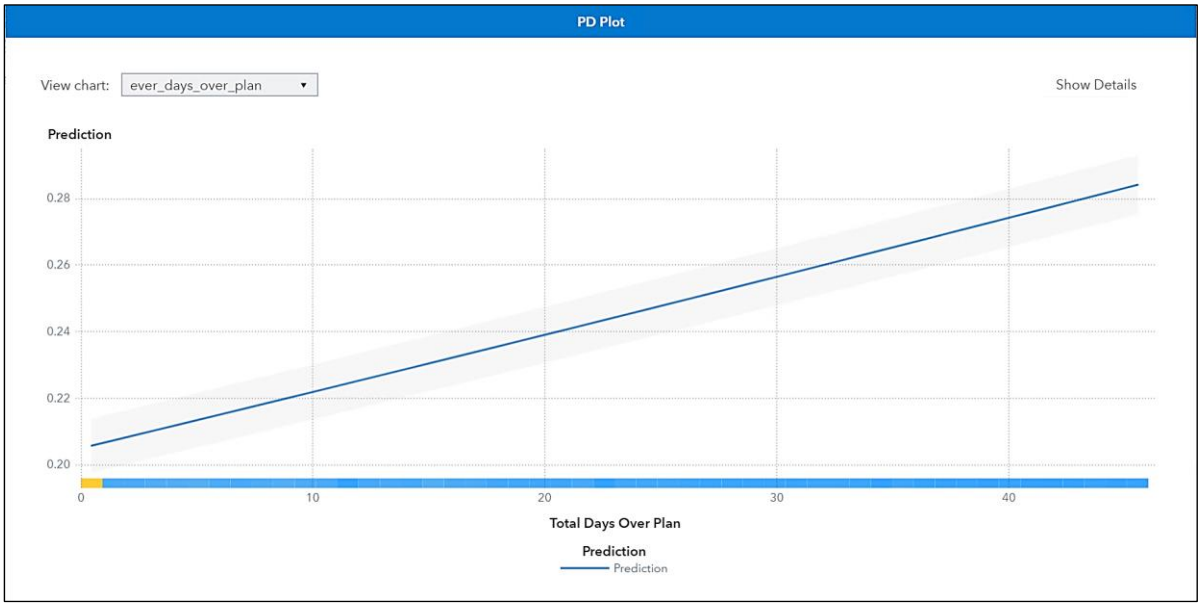
Surrogate Model Variable Importance

Variable Label	Variable Name	Relative Importance	Role	Variable Level
Total Days Over Plan	ever_days_over_plan	1	INPUT	INTERVAL
Handset Age Group	handset_age_grp	0.9146	INPUT	NOMINAL
Delinquent Indicator	delinq_indicator	0.6440	INPUT	NOMINAL
Transformed MB of Data Usage Month 6	LOG_MB_Data_Usg_M06	0.4131	INPUT	INTERVAL
Number of Times Suspended	times_susp	0.3311	INPUT	NOMINAL
Days Suspended Last 6M	avg_days_susp	0.3260	INPUT	INTERVAL
Open Work Orders	wrk_orders	0.2791	INPUT	NOMINAL
Total Late Payments Lifetime	pymts_late_ltd	0.2235	INPUT	NOMINAL
Imputed Replacement: Minutes On Network Pct	IMP_REP_MOU_ONNET_PCT_	0.0491	INPUT	INTERVAL

The most important inputs are listed in descending order of their importance. **Total Days Over Plan** appears to be the most important predictor, followed by **Handset Age Group**, and so on. Input relative importance is calculated by depth-one decision trees using each input to estimate the predicted values of the support vector machine model. PD and ICE plots use this table to choose the top predictors.

Note: If the trained model is a tree-based model (decision tree, forest, gradient boosting), PD and ICE uses the model variable importance table to choose the top predictors and both model variable importance and surrogate variable importance tables are provided in Model Interpretability results.

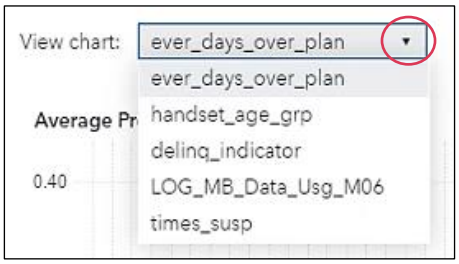
9. Expand the **PD Plot**.



The partial dependence plot shows the marginal effect of a feature on the predicted outcome of a previously fit model. The prediction function is fixed at a few values of the chosen features and averaged over the other features. A partial dependence plot can show whether the relationship between the target and a feature is linear, monotonic, or more complex. For interval inputs, the 95% confidence interval for the average target prediction is indicated by the shaded band around the line. The X axis includes a heat map that shows the distribution of **Total Days Over Plan**.

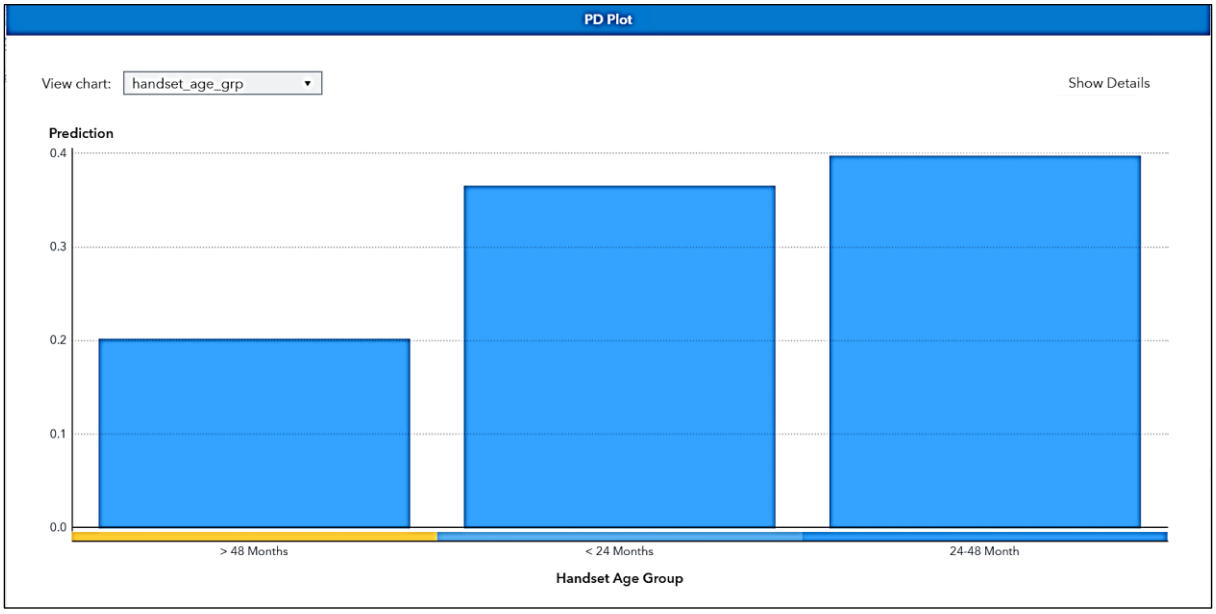
The plot above shows the relationship between **Total Days Over Plan** and the model's prediction. There is a positive linear relationship. On average, the more time (days) that a customer uses their device over their specified plan, there is an increase in the probability of churn. This is an important insight for the business.

10. To see the relationship between the model's predictions and other variables, click the drop-down arrow next to **View chart**.



These are the five most important inputs in the model, based on a one-level decision tree for all inputs used to predict the predicted values from the model.

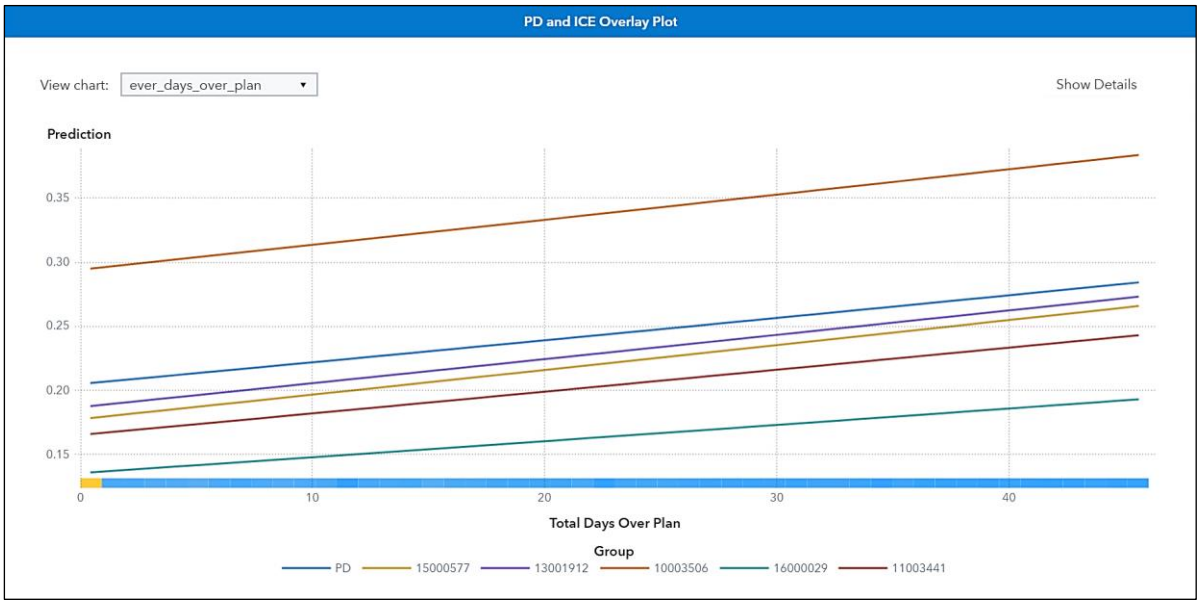
11. Select the **Handset Age Group** variable.



This PD plot indicates that there is a decrease in the probability of churn the older the handset is. Does this make business sense?

12. Exit the maximized view of the PD plot.

13. Expand the **PD and ICE Overlay Plot**.

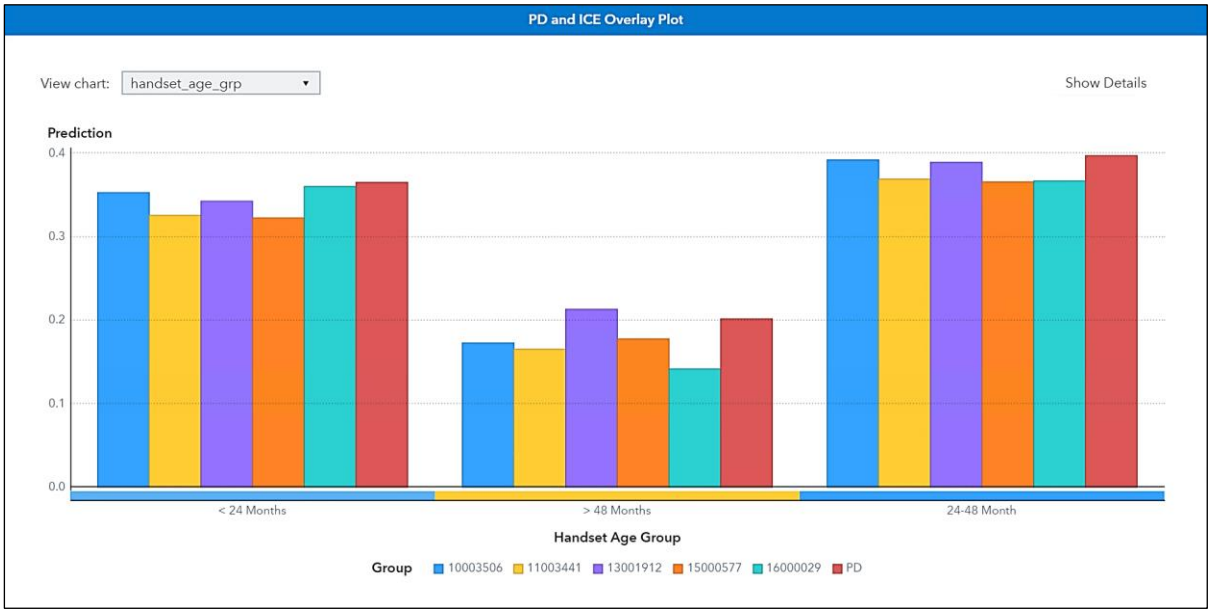


For a chosen feature, ICE plots draw one line per instance, representing how the instance's prediction changes when the feature changes. This plot shows the partial dependency (PD) and the relationship between total days over plan and the predicted target for each individual observation. For each individual observation, the corresponding ICE plot displays values of total days over plan on the X axis and the corresponding prediction for the target variable on the Y axis, holding the other inputs constant at their values for each observation.

The plot indicates that for all the five instances, there is a consistent increase in the probability of churn, for an increase in total days over plan, given that other features are constant.

ICE plots help resolve interesting subgroups and interactions between model variables. The most useful feature to observe when evaluating an ICE plot of an interval input is intersecting slopes. Intersecting slopes indicate that there is an interaction between the plot variable and one or more complementary variables. **Total Days Over Plan** does not show any interactions**.**

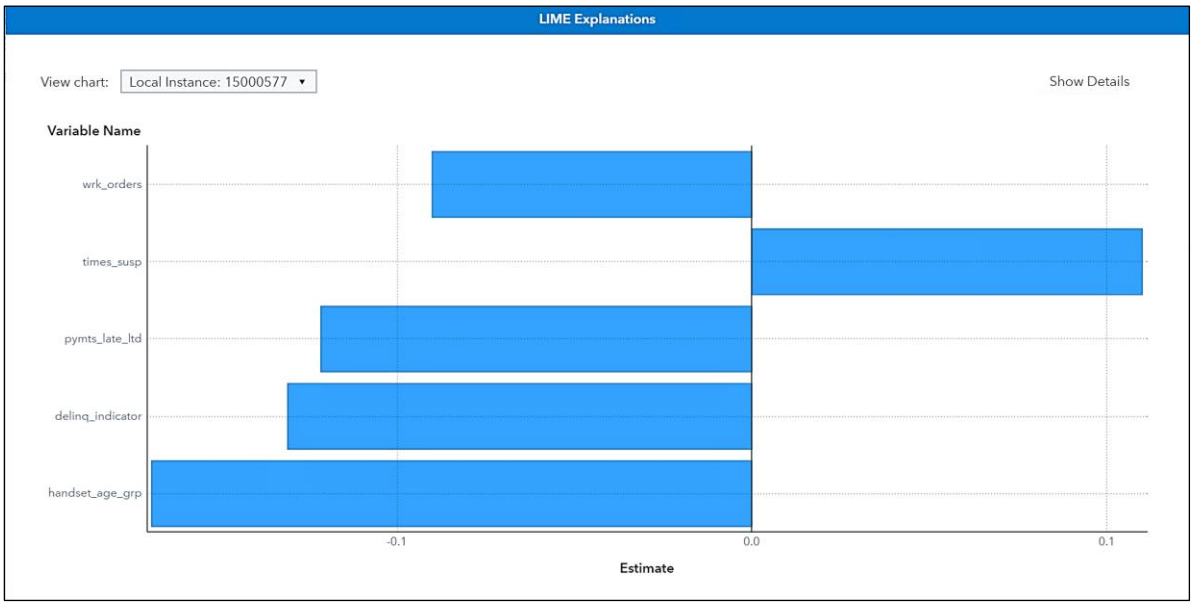
14. Click the drop-down arrow next to **View chart** to see ICE plots of other variables. Select the **Handset Age Group** variable.



This plot shows the partial dependency (PD) and the relationship between **Handset Age Group** and the predicted target for each individual observation. For each individual observation, the corresponding ICE plot displays values of **Handset Age Group** on the X axis and the corresponding prediction for the target variable on the Y axis, holding the other inputs constant at their values for each observation.

An ICE plot is created on individual observations. The most useful feature to observe when evaluating an ICE plot of a categorical input is significant differences between individuals in each group. Significant differences between each individual's plot between groups indicate group effects.

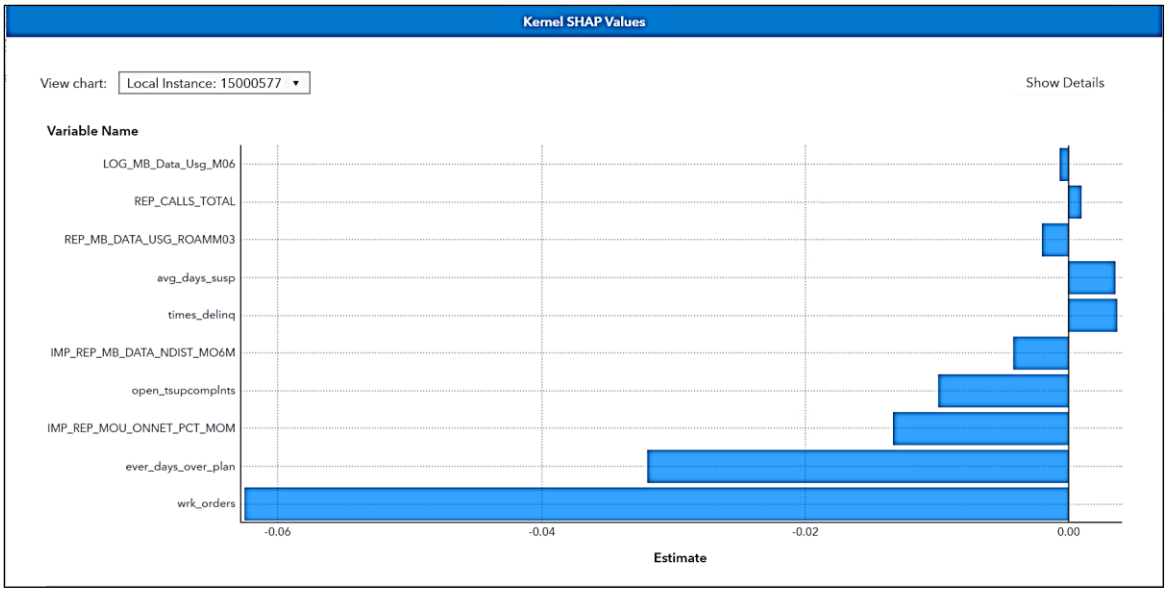
15. There are five individuals represented in this plot, with the average predicted probability of churn calculated separately for each individual, across all levels of **Handset Age Group**. For this variable, the trend of observing the lowest probability of churn for the oldest handset age group holds true for all the five individuals, and the average predicted probability of churn for the oldest handset group is slightly smaller in cluster 1 compared to cluster 2. In fact, for all levels of handset age group, the probability of churn in cluster 1 is slightly smaller than that for cluster 2.Exit the maximized view of the PD and ICE overlay plot. Although PD and ICE plots provide only indirect approximations of a model's workings, they are popular and highly visual tools for obtaining a working understanding of increasingly complicated machine learning models.
16. Expand the **LIME Explanations** plot.



This LIME plot displays the regression coefficient (estimate) for the inputs selected in the local surrogate linear regression model for fitting the predicted probability of the event "1" for the target churn for each individual observation. The inputs are ordered by significance in the chart, with the most significant input for the local regression model appearing at the bottom of the chart.

The LASSO technique is used to select the most significant effects from the set of inputs that was used to train the model. A positive estimate indicates that the observed value of the input increases the predicted probability of the event. For example, the value of " 0" for delinq_indicator decreases the predicted probability of the event "1" for the target churn by 0.13084 compared to the individual having a different value for delinq_indicator.

17. Restore the **LIME Explanations** plot.
18. Expand the **Kernel SHAP Values** plot.



For each individual observation, an input's Shapley value is the contribution of the observed value of the input to the predicted probability of the event "1" for the target churn. The Shapley values of all inputs sum to the predicted value. The inputs are displayed in the chart ordered by importance according to the absolute Kernel SHAP values.

The Kernel SHAP values are the regression coefficients that are obtained by fitting a weighted least squares regression. Note that each nominal input is binary-encoded based on whether it matches the individual observation. Interval inputs are binary encoded based on their proximity to the individual observation.

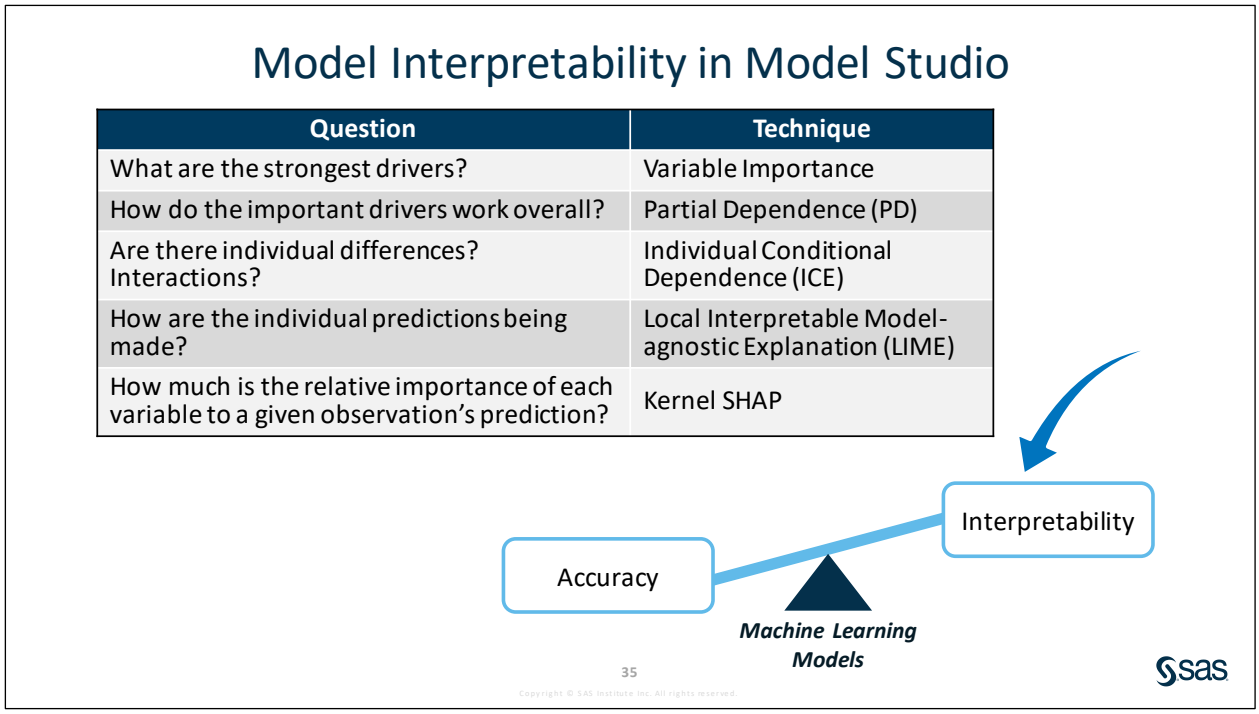
19. Restore the **Kernel SHAP Values** plot.
20. Close the Results window.
21. Run the entire pipeline and view the results of model comparison.

Champi...	Name	Algorith...	KS (You...	Misclas...
✓	SVM	SVM	0.5662	0.0729
	Logistic Regression	Logistic Regression	0.5488	0.0808

The support vector machine model is the champion of this pipeline based on default KS.

22. Close the Results window.

End of Demonstration



There is almost always a trade-off in terms of accuracy versus interpretability. Machine learning algorithms are good at generating very accurate (and generalisable) predictive models using quite complex combinations of mathematical and logical elements. They provide very good predictions, but it can be nearly impossible to understand how they arrived at those predictions or, in general, what the behaviour or trend of the model is. This is a big problem in regulated industries and other applications where it is important to explain ***why*** a model gave a certain answer. So, model interpretability (understanding the predictions) is the usual criticism of machine learning models. As machine learning models become more sophisticated, the ability to quickly and accurately interpret these models can diminish.

This criticism stems from the complex parameterisations found in the model. Although it is true that little insight can be gained by analysing the actual parameters of the model, much can be gained by analyzing the resulting prediction decisions. SAS Visual Data Mining and Machine Learning provides three plots that help users interpret model results:

Partial Dependence – A PD plot depicts the functional relationship between the model inputs and the model's predictions. A PD plot shows how the model's predictions partially depend on the values of the input variables of interest marginally when the predictions are averaged over other variables. To create a one-way PD plot, identify the plot variable. Next, a replica of the training data for each unique value of the plot variable will be created. The plot variable is replaced by the current unique value in each replicate. Finally, score each replicate with your model and compute the average predicted value within each replicate. The final result is a view of how the prediction changes with respect to the plot variable.

PD plots from various machine learning models can also be used for model comparison.

Individual Conditional Expectation – An ICE plot presents a disaggregation of the PD plot to reveal interactions and differences at the observation level. The ICE plot is generated by choosing a plot variable and replicating each observation for every unique value of the plot variable. Then, each replicate is scored. SAS Visual Data Mining and Machine Learning create a segmented ICE plot. A segmented ICE plot is created from a cluster of observations instead of individual observations.

Local Interpretable Model-Agnostic Explanations – The LIME method explains an individual prediction. This is done by fitting a linear regression model around that individual row based on a perturbed data set. This data set is based on the distribution of the original input data. The sample set is scored by the original model, and sample observations are weighted based on proximity to the observation of interest. Next, variable selection is performed using the LASSO technique. Finally, a linear regression is predicted to explain the relationship between the perturbed input data and the perturbed target variable. The final result is an easily interpreted linear regression model that is valid near the observation of interest.

Kernel SHAP—Generally speaking, Shapley values help you determine the relative importance of each variable to a given observation's prediction. In the feature space, Shapley values help you determine where you are, how you got there, and how influential each variable is at that location. This is in contrast to LIME values, which help you determine how changes in a variable's value affect the model's prediction. However, LIME and Shapley values should not be directly compared because they measure different behaviours.

Specifically, Kernel SHAP values are the parameter estimates of a weighted linear regression on the predicted probability of the selected observation. These values indicate the most influential variables for a selected observation compared against a reference data set, which is typically, though not necessarily, the training data.

Note: Each of these plots works for all models and is used to compare results across many different models.